Università degli studi di Bergamo

Scuola di Ingegneria (Dolmine)

CCS Ingegneria Edile

LM-24 Ingegneria delle Costruzioni Edili

Complementi di Scienza delle Costruzioni

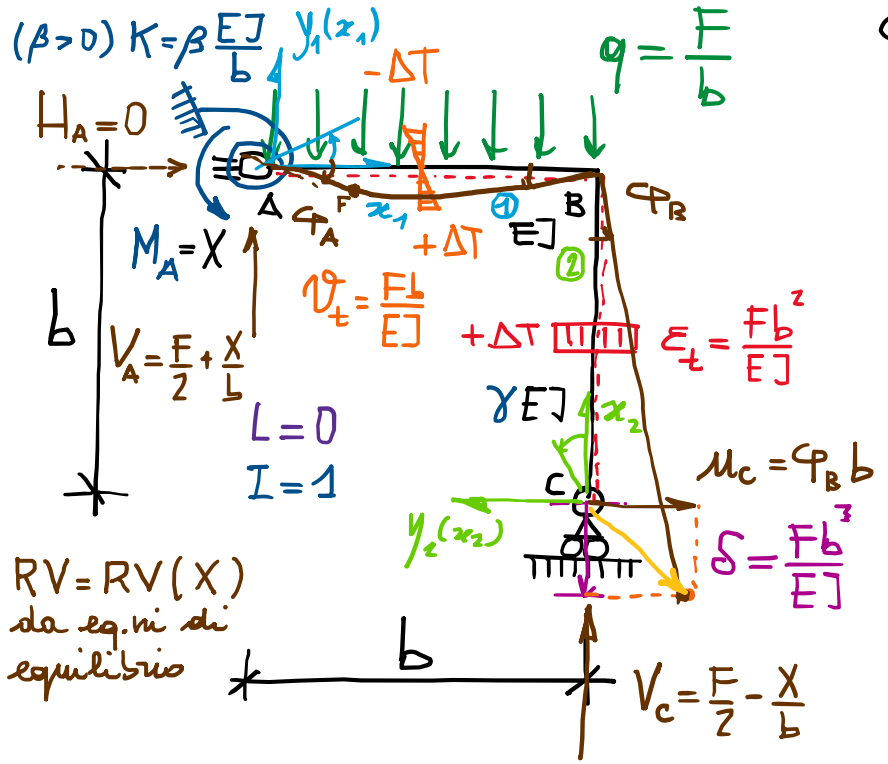
(ICAR/08 - SdC; 6 CFU)

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LEZIONE 09

Soluzione tramite Metodo della Linea Elastica (LE)



Curvatura totale del generico campo i-esimo: (n campi di integrazione)

$$y_i''(x_i) \simeq \chi_i(x_i) = \chi_{ie}(x_i) + \chi_{it}(x_i)$$

$|y_i'| \ll 1$

$\chi_{ie}(x_i) = \frac{M_i(x_i)}{EJ_i}$ — Legge di B-E-N

$\chi_{it}(x_i) = \vartheta_{it}(x_i)$ — curvatura termica impressa

$$\gamma_i(x_i)\,EJ \quad \text{con } \gamma_i(x_i) = EJ_i(x_i)$$

$$EJ_i(x_i)\, y_i''(x_i) = M_i(x_i) + EJ_i(x_i)\,\vartheta_{it}(x_i)$$

$$\boxed{EJ \cdot y_i''(x_i) = \frac{M_i(x_i)}{\gamma_i(x_i)} + EJ \cdot \vartheta_t(x_i)}$$

eq. differenziale del 2° ordine

Tratto 1:

$$EJ\, y_1''(x_1) = M_1(x_1) + EJ\,\frac{Fb}{EJ} = -\frac{F}{b}\frac{x_1^2}{2} + \underbrace{\left(\frac{F}{2}+\frac{X}{b}\right)}_{V_A} x_1 - X + Fb$$

$\int dx_i$

$$EJ\, y_1'(x_1) = -\frac{F}{b}\frac{x_1^3}{6} + \left(\frac{F}{2}+\frac{X}{b}\right)\frac{x_1^2}{2} + (Fb - X)x_1 + A_1$$

$\int dx_i$

$$EJ\, y_1(x_1) = -\frac{F}{b}\frac{x_1^4}{24} + \left(\frac{F}{2}+\frac{X}{b}\right)\frac{x_1^3}{6} + (Fb - X)\frac{x_1^2}{2} + \underbrace{A_1 x_1 + A_2}_{\text{comp. moto rigido rototraslatorio}}$$

Tratto 2:

$$EJ\, y_2''(x_2) = \frac{0}{\gamma} + EJ\,0 = 0$$

$$EJ\, y_2'(x_2) = B_1$$

$$EJ\, y_2(x_2) = \underbrace{B_1 x_2 + B_2}$$

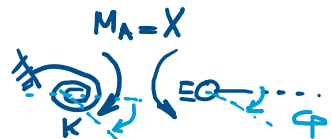
Incognite $2n + I = 5$

A_1, A_2, B_1, B_2; X

costanti di integrazione — inc. iperst.

Scrittura e imposizione delle condizioni al contorno (c.c.) [in numero pari a $n_{cc} = 2n + I = 5$]

- $y_1(0) = 0$ — cerniera in A
- $y_1'(0) = -\frac{X}{K}$ — molla elastica rotazionale in A
- $y_1(b) = -\delta + \varepsilon_t b = 0$ — carrello in C con cedimento $\delta + \left(\Delta\ell_t^{BC} = \varepsilon_t b\right)$
- $y_1'(b) = y_2'(b)$ — continuità alla rotazione in B
- $y_2(b) = 0$ — cerniera in A + inest. assiale di AB ($EA_{AB} \to \infty$)



$\varphi = \frac{X}{K}$ opposta a $M_A = X$ sull'asta

$\rightarrow$ A φ_B

spost. assoluti

B è nodo fisso ($u_B = v_B = 0$)

$EJ\ y_1(0) = 0 \Rightarrow \boxed{A_2 = 0}$

$EJ\ y_1'(0) = A_1 = -\frac{X}{K} EJ = -\frac{X}{\beta \frac{EJ}{b}} EJ = -\frac{b}{\beta} X \Rightarrow \boxed{A_1 = A_1(X) = -\frac{b}{\beta} X}$

$EJ\ y_1(b) = -\frac{F}{b}\frac{b^4}{24} + \left(\frac{F}{2} + \frac{X}{b}\right)\frac{b^3}{6} + (Fb - X)\frac{b^2}{2} + A_1 b + A_2 = 0$

$EJ\ y_1'(b) = -\frac{F}{b}\frac{b^3}{6} + \left(\frac{F}{2} + \frac{X}{b}\right)\frac{b^2}{2} + (Fb - X)b + A_1 = EJ\, y_2'(b) = B_1 =$

$EJ\ y_2(b) = B_1 b + B_2 = 0 \Rightarrow B_2 = -B_1 b \rightarrow u_C = \varphi_B b \quad (v_C = -\delta)$

$\varphi_B \geq |\varphi_A| \qquad \beta \geq 0$

$\varphi_B = \frac{B_1}{EJ} = \frac{13}{8}\frac{1+\beta/6}{3+\beta}\frac{Fb^2}{EJ}$

$\boxed{B_2 = -\frac{13}{48}\frac{6+\beta}{3+\beta} Fb^3}$

$\boxed{B_1 = \frac{13}{48}\frac{6+\beta}{3+\beta} Fb^2}$

$\circlearrowleft \varphi_A = \frac{A_1}{EJ}$

$\rightarrow A_1 = -\frac{b}{\beta}\frac{13}{8}\frac{\beta}{3+\beta} Fb = \boxed{-\frac{13}{8}\frac{1}{3+\beta} Fb^2 = A_1}$

$\Rightarrow \left(\frac{1}{6} - \frac{31}{32} - \frac{1}{\beta}\right) Xb^2 = \left(\frac{1}{24} - \frac{1}{12} - \frac{1}{2}\right) Fb^3$

$\left(-\frac{1}{3} - \frac{1}{\beta}\right) X = \frac{1-2-12}{24} Fb = -\frac{13}{24} Fb$

$\frac{3+\beta}{\beta} X = \frac{13}{24_8} Fb$

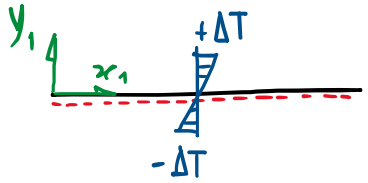
$\boxed{X = \frac{13}{8}\frac{\beta}{3+\beta} Fb}$

$\beta \to 0,\ X \to 0$

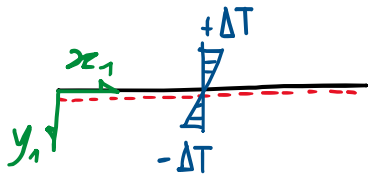
$\beta \to \infty,\ X \to \frac{13}{8} Fb$

– Le finali, sost. A_1, A_2, B_1, B_2 e X e tracciamento della deformata qualitativa –

- Attenzione ai segni su LE e e.c.!

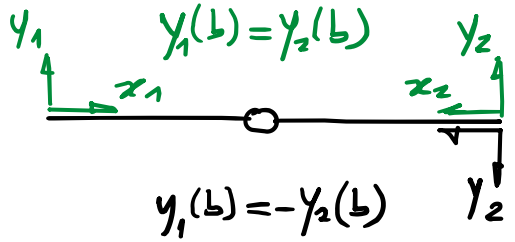


$EJ y_1'' = +M - EJ\vartheta_t$

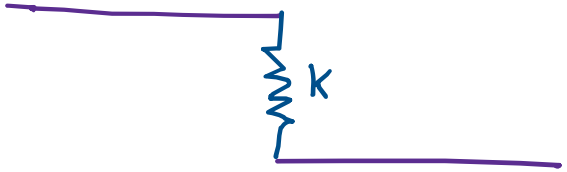


$EJ y_1'' = -M + EJ\vartheta_t$

curvature elastiche e termiche



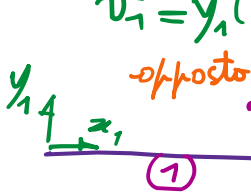
molla relativa elongazionale



letto nel riferim. locale

$\vartheta_1 = y_1(b)$

opposto a X

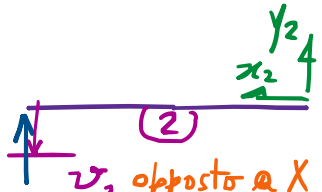




spost. relativo

$\Delta l = \vartheta_{rel} = \vartheta_1 + \vartheta_2$

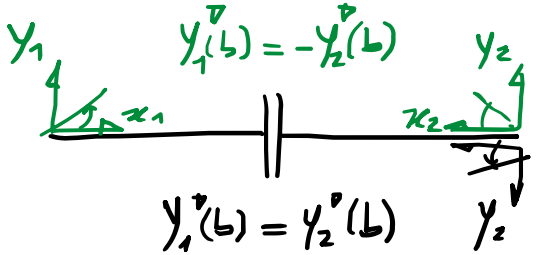
$= \frac{X}{K}$ legge di Hooke



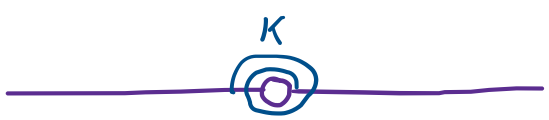
ϑ_2 opposto a X

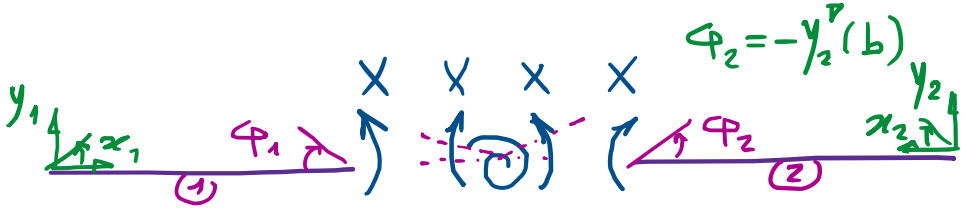
X $\vartheta_2 = -y_2(b)$

$$y_1(b) - y_2(b) = \frac{X}{K}$$



rotazionale





$\varphi_1 = -y_1'(b)$

$$-y_1'(b) - y_2'(b) = \frac{X}{K}$$

$\Delta\varphi = \varphi_{rel} = \varphi_1 + \varphi_2$

$= \frac{X}{K}$